

Data Analyst Capstone project

Case Study : How Can a Wellness Technology Company Play It Smart?

OVERVIEW

Urška Sršen and Sando Mur founded Bellabeat, a high-tech company that manufactures health-focused smart products. Collecting data on activity, sleep, stress, and reproductive health has allowed Bellabeat to empower women with knowledge about their own health and habits. Sršen knows that an analysis of Bellabeat's available consumer data would reveal more opportunities for growth. She has asked the marketing analytics team to focus on a Bellabeat product and analyze smart device usage data in order to gain insight into how people are already using their smart devices. Then, using this information, she would like high-level recommendations for how these trends can inform Bellabeat marketing strategy.

GOAL/PURPOSE

To analyze the usage trends of smart device(s) of Bellabeat and to implement action plans like marketing strategies on less used (sold) device(s) and further improvements in frequently used devices based on the insights gathered from the analysis.

Bellabeat has several products, but for this analysis, we will focus on the application. The Bellabeat app provides users with health data related to their activity, sleep, stress, menstrual cycle, and mindfulness habits. This data can help users understand their current practices and make healthy decisions.

ANALYSIS STEPS

1. ASK

Business Task: To identify the usage trends of smart devices and insights from this analysis should provide high-level recommendations for how these trends can inform Bellabeat marketing strategy.

The key stakeholders include:

- Urška Sršen — Chief Creative Officer and Bellabeat's Co-founder.
- Sando Mur — Mathematician and Bellabeat's Co-founder.
- Bellabeat's marketing analytics team — a team of data analysts.

Below are the SMART questions to be addressed for the analysis.

Specific	Measurable	Action-Oriented	Relevant	Time-bound
What are the primary business goals you hope to achieve through this analysis of wellness technology growth opportunities?	What are the key performance indicators currently used to measure the success of our wellness technology products?	What are the specific areas of growth (new product features or user acquisition) which need to be explored?	What are the current revenue streams associated with our wellness products?	Are there any specific growth targets or timelines we should be aware of?

2. PREPARE

The dataset is public, accessible and made available in Kaggle. It is licensed under CC0: Public Domain. It is generated by respondents to a distributed survey via Amazon Mechanical Turk between 03.12.2016-05.12.2016. **Thirty** eligible Fitbit users **consented** to the submission of personal tracker data, including minute-level output for physical activity, heart rate, and sleep monitoring. The dataset is a zip file dated and upon extract could see 18 csv files each corresponding to different activities like heartbeat, hours of sleep, steps and so on. Below are my findings about the dataset.

The data is in long format.

Some of the column names are inconsistent and less clear.

The data has only 30 users' records and that too a month's record. It didn't represent a complete population. There are no details like age, gender, address or other information about the customer. Hence there could be a possibility of sampling bias.

The data is outdated as it has records of 2016.

Reliable	Original	Comprehensive	Current	Cited
Accurate but incomplete and possibility of bias	Data collected directly from customers through survey posted in trusted public website like Kaggle	Has most information required for analysing usage trends of fitness product	Not the recent data	Yes

3. PROCESS

I'm using Google Sheets for cleaning the data as the number of user records are around 30 and can be used to generate simple charts. Imported the csv files into google sheets as a new spreadsheet named "**FitBit Fitness Tracker Dataset**". I have used 10 data tables out of 18 for analysis which are mentioned below.

sleepDay_merged.csv

weightLogInfo_merged.csv

hourlyCalories_merged.csv

hourlyIntensities_merged.csv

dailyActivity_merged.csv

dailyCalories_merged.csv

dailyIntensities_merged.csv

dailySteps_merged.csv

minuteSleep_merged.csv

hourlySteps_merged.csv

1. Adjusted the column and row height for a clear view of contents.
2. Standardized the column names with spaces and bolding the fonts. Added units of measurements for clear understanding.
3. Checked for blanks or null or N/A rows by using filter option. In the "minutesleep_merged" sheet there were too many rows blank which were removed. There were many blank values in the "Fat" column of the "WeightLogInfo_merged" sheet which needed to be collected from an appropriate source later and hence removed for current analysis.
4. Looked for any duplicate rows and removed them using the Data cleanup option. In the "minutesleep_merged" sheet 543 duplicates were removed and 187978 remained while in the "sleepday_merged" sheet 3 duplicates were removed and 410 unique values remained.
5. Formatted the date and date-time columns in all sheets as Date and DateTime respectively.
6. Formatted the distance, BMI, weight and intensity columns by rounding up to decimal places.
7. Checked for spelling mistakes and misfielded values.
8. Checked for extra spaces using TRIM functions.

4. ANALYZE

Firstly to identify the trends and patterns in all sheets. Number of IDs in each data sheet is identified by selecting the filter option in each ID column and getting the total number of IDs and are as follows.

sleepDay_merged - 24

weightLogInfo_merged - 2

hourlyCalories_merged - 33

hourlyIntensities_merged - 33

dailyActivity_merged - 33

dailyCalories_merged - 33

dailyIntensities_merged - 33

dailySteps_merged - 33

minuteSleep_merged - 24

hourlySteps_merged - 33

Therefore we will focus on data sheets which contain 33 user data for eliminating sampling bias, ensuring consistency.

hourlyCalories_merged - 33

hourlyIntensities_merged - 33

dailyActivity_merged - 33

dailyCalories_merged - 33

dailyIntensities_merged - 33

dailySteps_merged - 33

hourlySteps_merged - 33

As a second step, we can identify the trends and patterns in the above data sheets. For this we can use SQL in bigquery. Uploaded the above cleaned csv files into BigQuery datasets for analysis.

Identified that the tables related to daily activities like DailyActivity, DailyIntensities, DailySteps, DailyCalories, all have 940 rows with the ID as primary key and activity days between the dates 2016-04-12 and 2016-05-12. Also columns in Calories, Intensities and Steps tables are aggregated in the Daily Activity table.

Confirmed this pattern by running JOIN Query in all three tables with the DailyActivity table.

Untitled query [Run](#) [Save](#) [Download](#) [Share](#) [Schedule](#) [Open in](#)

```

1 SELECT act.Id, act.ActivityDate, cal.ActivityDay, TotalSteps, cal.Calories FROM `capable-vista-456414-k4.FitBit_Fitness_Tracker_Nara.DailyActivity` act
2 FULL OUTER JOIN `capable-vista-456414-k4.FitBit_Fitness_Tracker_Nara.DailyCalories` cal
3 ON act.Id = cal.Id
4 AND act.ActivityDate = cal.ActivityDay

```

Query completed

Query results [Save results](#)

Job information [Results](#) [Chart](#) [JSON](#) [Execution details](#) [Execution graph](#)

row	Id	ActivityDate	ActivityDay	TotalSteps	Calories
1	3977333714	2016-04-16	2016-04-16	13459	1625
2	3977333714	2016-05-07	2016-05-07	11550	1574
3	5577150313	2016-04-21	2016-04-21	10830	4018
4	8583815059	2016-04-22	2016-04-22	8687	2944
5	1844505072	2016-04-23	2016-04-23	3570	1645
6	1927972279	2016-04-15	2016-04-15	980	2221
7	6290855005	2016-04-20	2016-04-20	6361	2701

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I have used Outer Join to identify if there are any unmatched values in both the tables. And based on the query results there are no unmatching values in both activity and calories tables and the total number of rows matched to 940. Similar approach was done for steps and intensities tables.

Untitled query

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```
1 SELECT act.Id, act.ActivityDate, int.ActivityDay, TotalSteps, int.VeryActiveDistance FROM `capable-vista-456414-k4.FitBit_Fitness_Tracker_Nara.DailyActivity` act
2 FULL OUTER JOIN `capable-vista-456414-k4.FitBit_Fitness_Tracker_Nara.DailyIntensities` int
3 ON act.Id = int.Id
4 AND act.ActivityDate = int.ActivityDay
```

Query completed

Query results

Save resultsOpen

Job informationResultsChartJSONExecution detailsExecution graph							
Row	Id	ActivityDate	ActivityDay	TotalSteps	VeryActiveDistance		
1	8583815059	2016-05-11	2016-05-11	8701	0.37000000476...		
2	2347167796	2016-04-24	2016-04-24	9471	0.0		
3	4445114986	2016-04-28	2016-04-28	4493	0.0		
4	4445114986	2016-05-04	2016-05-04	2923	0.0		
5	5553957443	2016-04-16	2016-04-16	5771	0.0		
6	8792009665	2016-05-02	2016-05-02	1831	0.0		
7	4702921684	2016-05-02	2016-05-02	7245	0.37999999523...		

Results per page: 501 – 50 of 940

Untitled query

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```
1 SELECT act.Id, act.ActivityDate, act.TotalSteps, steps.StepTotal FROM `capable-vista-456414-k4.FitBit_Fitness_Tracker_Nara.DailyActivity` act
2 FULL OUTER JOIN `capable-vista-456414-k4.FitBit_Fitness_Tracker_Nara.DailySteps` steps
3 ON act.Id = steps.Id
4 AND act.ActivityDate = steps.ActivityDay
```

Query completed

Query results

Save results

Job informationResultsChartJSONExecution detailsExecution graph					
Row	Id	ActivityDate	TotalSteps	StepTotal	
1	4702921684	2016-04-22	1664	1664	
2	5553957443	2016-05-01	5164	5164	
3	1927972279	2016-04-24	2090	2090	
4	7007744171	2016-04-12	14172	14172	
5	4558609924	2016-05-03	5267	5267	
6	8877689391	2016-04-14	21129	21129	
7	1503960366	2016-04-15	9762	9762	

Results per page: 501 – 50 of 940

Based on the above query results, it is concluded that daily calories, daily intensities, and daily steps tables have the same data as the daily activity table. Hence we can consider only the daily activity table. Now we have only

dailyActivity_merged - 33

hourlyCalories_merged - 33

hourlyIntensities_merged - 33

hourlySteps_merged - 33

Checked the consistency of all 3 hourly tables and are consistent having 22099 rows with unique users, 24hrs data in each day throughout the month. Confirmed this by running the below query in BigQuery.

```
SELECT count(*) FROM
`capable-vista-456414-k4.FitBit_Fitness_Tracker_Nara.HourlyCalories`;

SELECT count(*) FROM
`capable-vista-456414-k4.FitBit_Fitness_Tracker_Nara.HourlyIntensities`;

SELECT count(*) FROM
`capable-vista-456414-k4.FitBit_Fitness_Tracker_Nara.HourlySteps`;
```

Then we need to find user engagement by checking the daily activity table. We can do this by analyzing how many records are there for each user in the daily activity table.

```
SELECT Id, count(Id) as total_activities_records FROM
`capable-vista-456414-k4.FitBit_Fitness_Tracker_Nara.DailyActivity`

group by Id

order by total_activities_records desc;
```

From the above query results it is implied that users with a minimum of 4 days and maximum of 31 days are actively logged in. Then ran the below query to find user engagement.

```
SELECT Id, count(Id) as total_activities_records,
CASE
WHEN count(Id) > 0 and count(Id) <=10 then "Less Active"
WHEN count(Id) >10 and count(Id) <=20 then "Moderately Active"
WHEN count(Id) >20 and count(Id)<=31 then "Highly Active"
END AS Active_Status
FROM `capable-vista-456414-k4.FitBit_Fitness_Tracker_Nara.DailyActivity`
group by Id order by Id;
```

5. SHARE

INSIGHT 1:

Based on the above query results, it is inferred that out of 33 users, 29 users are highly active, 3 users are moderately active and 1 user is less active.

INSIGHT 2:

Used pivot table to find out which day of the week, users stepped more on an average and it turns out to be on Saturdays, then consistently on weekdays and very less on Sundays. This insight would be useful for framing marketing strategy.

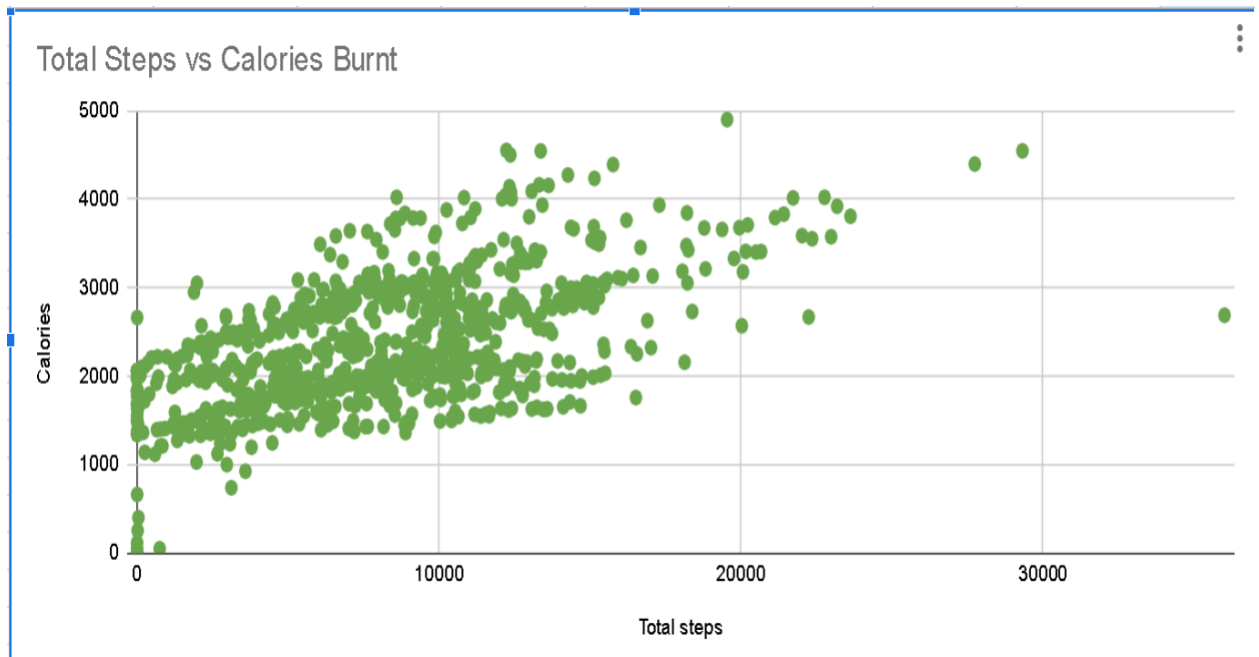
A	B	C
Activity Date - Day of the week	SUM of Total Steps	AVERAGE of Total Steps
Sunday	838921	6933.231405
Monday	933704	7780.866667
Tuesday	1235001	8125.006579
Wednesday	1133906	7559.373333
Thursday	1088658	7405.836735
Friday	938477	7448.230159
Saturday	1010969	8152.975806

Activity Date - Day of the week	AVERAGE of Total Steps	AVERAGE of Total Distance	AVERAGE of Calories
Sunday	6933.231405	5.03	2263
Monday	7780.866667	5.55	2324.208333
Tuesday	8125.006579	5.83	2356.013158
Wednesday	7559.373333	5.49	2302.62
Thursday	7405.836735	5.31	2199.571429
Friday	7448.230159	5.31	2331.785714
Saturday	8152.975806	5.85	2354.967742

Hence the total steps, distance and calories are correlated from the above pivot data.

INSIGHT 3:

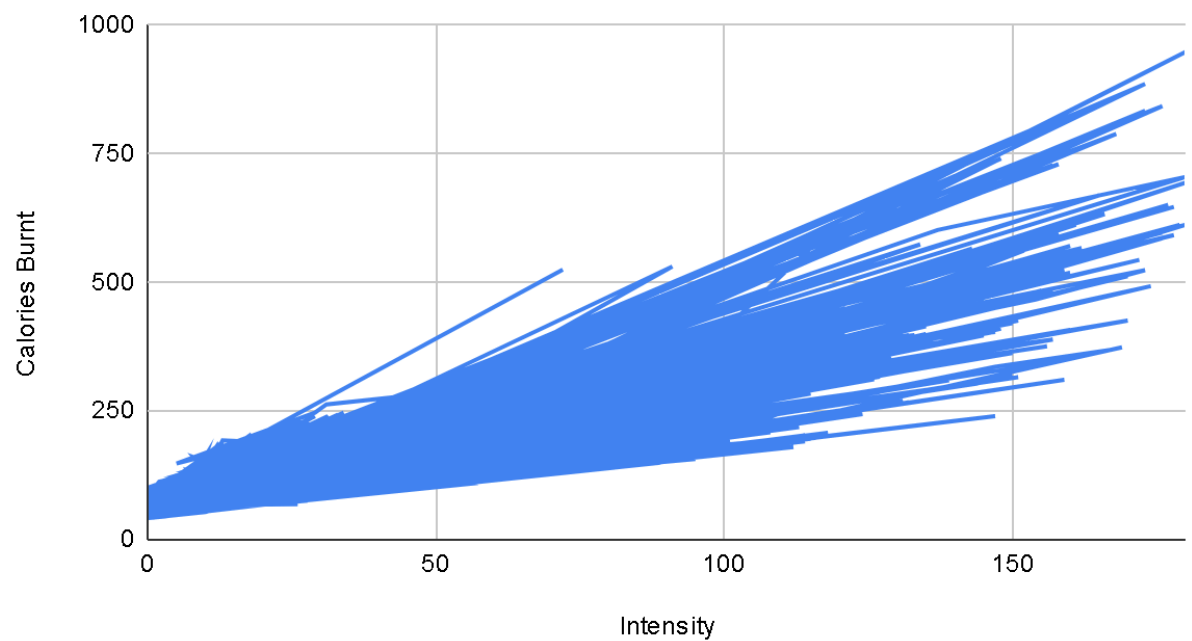
For visualization, I used Tableau as the number of rows are higher and spreadsheets are not effective for charts. Compared calories burnt and total steps data and inferred that there is slightly a positive correlation between these variables. Hence we can frame a strategy to encourage users to walk more steps to burn calories.



INSIGHT 4:

Intensity vs Calories chart also shows positive correlation.

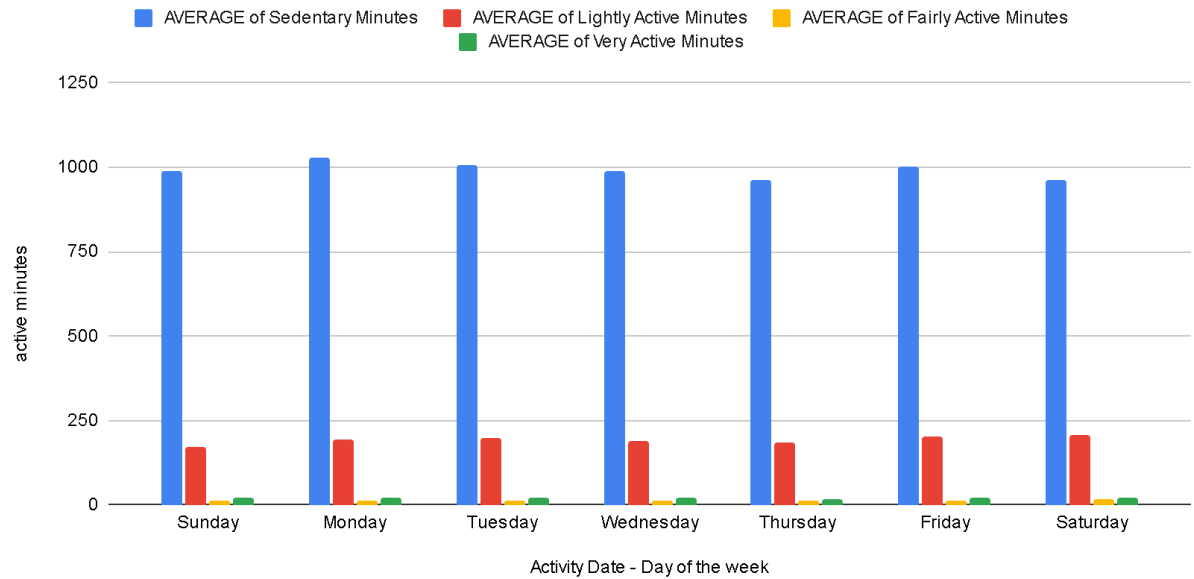
Intensity vs Calories Burnt



INSIGHT 5:

Activity levels based on the days of the week. Below chart shows there is little to no correlation between both the variables.

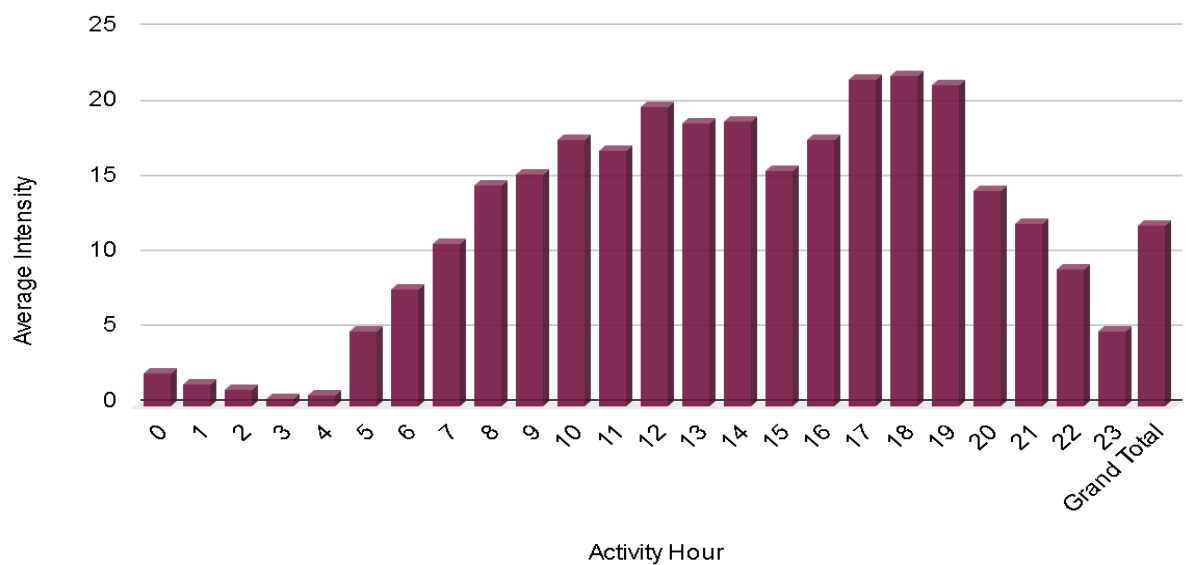
Average Active minutes vs Day of week



INSIGHT 6:

Average Intensity vs Active Hour

Average Intensity vs. Activity Hour



From the above chart it is observed that hours from 5 PM to 7 PM and 12 PM are the most active. This implies that users average intensity increases during after work hours and at lunch hour.

6. ACT

Below are my recommendations based on gathered insights.

1. Marketing strategies can be applied on Saturdays and weekdays as people walk more on these days. Also promotion or discounts can be given on Sundays to encourage people to workout on Sundays too.
2. Reminders can be set for each activity by making a threshold or to reach calorie goals set by the customers which are beneficial for both customers and business.
3. There is no data collected on women's menstruation cycle and that can be added as a tracker for effective workout management and to engage more female participants.
4. Since the number of steps is correlational to calories burnt, it can be added as a point to earn coupons/gift cards for achieving step count goals. This inturn increases customer engagement.

PREPARED BY

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